Essays

|!|BCEEC0140000001|!|

Outline the process from sperm production to fertilization. (10 marks)

##

Content	Description	Marks
Sperm production	- In the testis, cells in the walls of the seminiferous tubules carry out meiotic cell divisions to produce haploid sperms. (1) - Sperm cells mature in the epididymis and develop tadpole-like structures. (1) - Newly-formed sperms are temporarily stored in the epididymis. (1)	(Max. 2)
Ejaculation	- Sperm is activated and nourished by seminal fluid. (1) - Seminal fluid is secreted by seminal vesicles, Cowper's glands and prostate glands. Sperms are mixed with seminal fluid to form semen. (1) - When a male is sexually excited, the normally soft penis becomes erect and hard. (1) - During sexual intercourse, the erect penis of the male is inserted into the vagina of the female. (1) - Semen is ejaculated from the penis into the vagina near the cervix. (1)	(Max. 3)
Fertilization	- Sperms swim through the cervix, the uterus and into the oviducts. (1) - Sperms meet an ovum in the oviduct. (1) - A sperm penetrates the ovum by releasing enzymes to digest the jelly coat of the ovum. (1) - The nucleus of the sperm fuses with the nucleus of the ovum to form a zygote. (1)	(Max. 2)
Communication		(Max. 3)

(10 marks)

##

Skill assessed	Marks
Understanding basic concepts	7
Communication	3

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Describe the structure and functions of the placenta in the female's body. (12 marks)

##

Structure and adaptation: (Max. 6)

- The placenta is a disc-like organ formed in the uterus at the site of implantation. (1)
- The placenta has numerous villi which contain the embryo's capillaries. (1)
- This provides a large surface area for the exchange of materials. (1)
- The embryo's blood is close to the maternal blood. This reduces the distance for diffusion of materials. (1)
- The embryo's blood is separated from the material blood. (1)

  Any ***two*** of the following: (Max. 2)

  - This prevents the clumping of the embryo's blood and maternal blood when their blood groups are incompatible. (1)
  - This prevents the high blood pressure of the maternal blood from damaging the embryo's blood vessels. (1)
  - It prevents large pathogen and toxins in the maternal blood from entering the embryo's blood. (1)

Function: (Max. 3)

- In the placenta, nutrients, oxygen and antibodies diffuse from the maternal blood to the embryo's blood. (1)
- Metabolic wastes such as urea and carbon dioxide produced by the embryo diffuse from the embryo's blood to the maternal blood. (1)
- The placenta secretes hormones that help maintain the thickness of the uterine lining during pregnancy. (1)

Communication (Max. 3)

(12 marks)

##

Skill assessed	Marks
Understanding basic concepts	9
Communication	3

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|!|BCEEB0140000001|!| ★ **Cross topic**

Human adults and foetuses employ the lungs and the placenta respectively for gas exchange. These organs share some common principles in their structural adaptations to be effective. Describe how these common principles enable the two organs to work effectively for gas exchange. (11 marks)

##

Content: (Max. 8)

Both organs have a large surface area (1) for rapid gas exchange by diffusion:

- lungs: the presence of numerous air sacs (1)
- placenta: the presence of numerous chorionic villi (1)

The walls of air sacs, chorionic villi and capillaries are one-cell thick / very thin. (1)
The capillaries are in close contact with air sacs and chorionic villi. (1)
This shortens the diffusion distance (1) for rapid gas exchange by diffusion.

Both organs are highly vascularized / with numerous and dense capillary network (1) so that the absorbed gases can be transported away quickly (1) to maintain steep diffusion gradients of gases (1) for rapid gas exchange by diffusion.

Communication (Max. 3)

(11 marks)

##

Skill assessed	Marks
Understanding basic concepts	8
Communication	3

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